

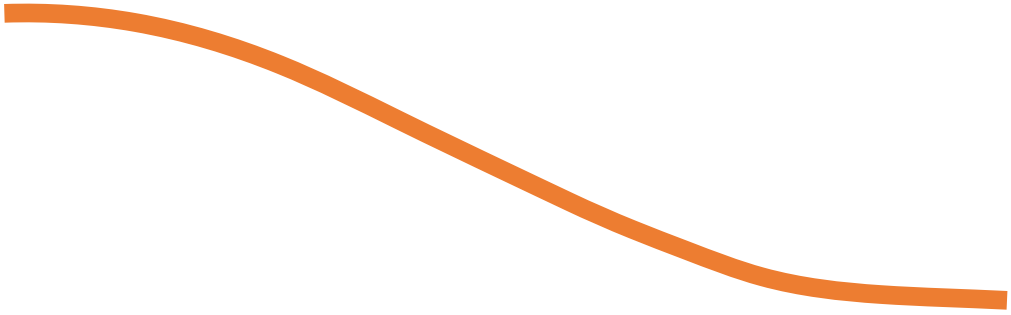






























Atmospheric Neutrino Problem



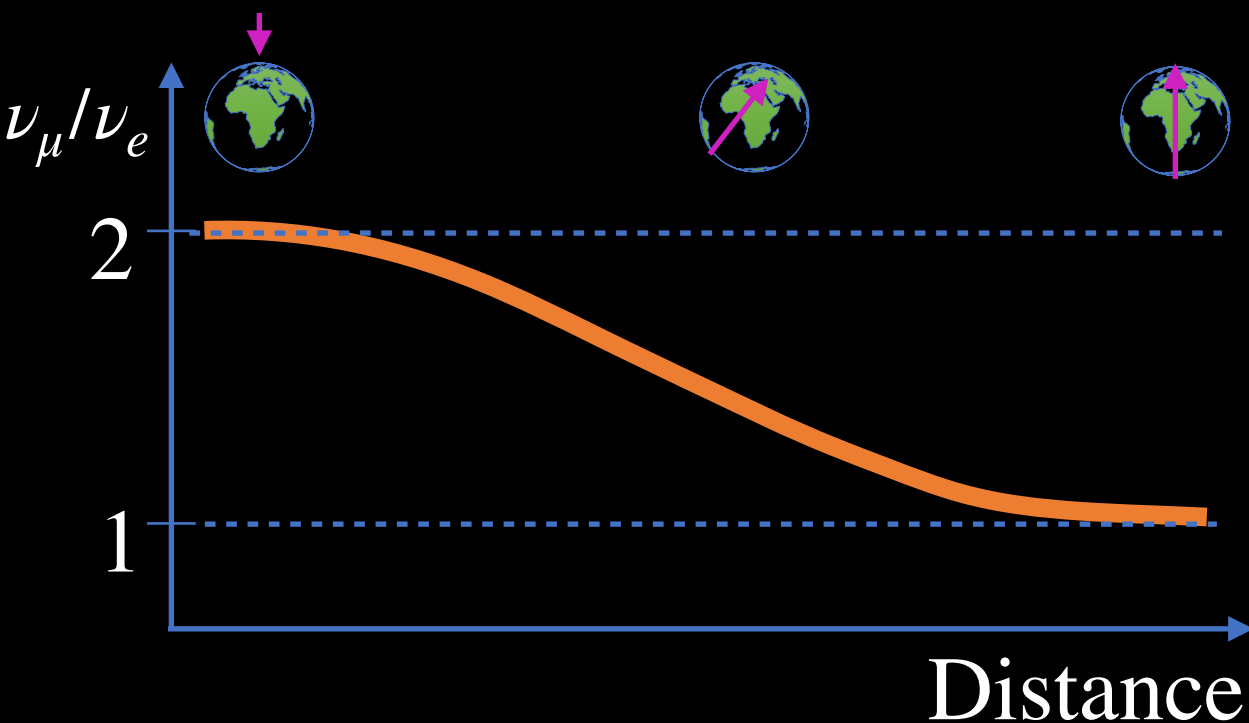


Atmospheric Neutrino

- 🌍 Earth is bombarded by cosmic rays
 - In inelastic collisions $p + A \rightarrow \text{hadrons} \approx \pi^\pm$
 - Pions decay: $\pi^+ \rightarrow \mu^+ + \nu_\mu$
 - Muons decay too: $\mu^+ \rightarrow e^+ + \nu_e + \bar{\nu}_\mu$



- 🌍 For each couple of muon neutrino one expects one electron neutrino



😲 Experiments discovered that ratio ν_μ/ν_e does not agree with expectation and depends upon neutrino travelled distance!

Atmospheric Neutrino Problem

6. Neutrino. Unknowns